Mathew Terhune

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EDUCATION

University of Victoria — Sep. 2021 – April 2025

BSc in Computer Science – 7.3/9 GPA — *Victoria, BC, Canada*

PROFESSIONAL EXPERIENCE

Cortex Labs: DECISION-MAP: WEBSITE — Sep. 2023 – April 2025

Undergraduate research assistant & developer — *Victoria, BC, Canada*

- Designed and helped maintain a visualization tool relating to brain injury, mental health, and addiction.
- Project has been presented globally at events including Canadian MP, Canadian Traumatic Brain Injury Research Consortium, and the Global Neuropsychology Congress in Portugal.

Columbia Basin Trust — May 2024 – Sep. 2024

CO-OP, Information Systems — *Castlegar, BC, Canada*

- Responded to and resolved support tickets, ensuring timely and effective troubleshooting of technical issues.
- Researched, tested, documented, presented, and prototyped the integration of Microsoft Power BI to management.
- Assisted is regular servicing and secure data wiping on physical servers.
- Managed and maintained multiple virtual machines, creating a virtual network for communication.

HIGHLIGHTED PROJECTS

Tangle Defence – Nelson 48-hour Game Jam: GitHub — July 2025

- Designed and developed a tower-defense game prototype in Godot using GDScript under strict time constraints.
- Implemented edge-based tower connection mechanism, wave & round system with enemy pathing, health and money system.
- Create custom parallax background, state management system, with in game economy.

Graph Search Algorithm Visualization Tool: GitHub — May 2025

- Developed interactive graph visualization tool using React, allowing real-time exploration of algorithms such as BFS & DFS
- Implemented dynamic grid resizing, node, selection, and visualization controls optimized with React state management and event handling.

Search Algorithm Comparison in Decentralized P2P Networks: Website | GitHub — Spring 2025

- Designed and implemented a custom Gnutella-like network simulation using C++ NS3 to evaluate search algorithms.
- Compared flooding, normalized flooding, and k-random walks across several different graph types.
- Created custom packet headers, topology, builders, and forwarding logic to simulate real life node behaviour and track metrics.

Parallel Pathfinding with A-Star: GitHub — Fall 2024

- Designed a GPU-optimized priority queue with CUDA to be used within a parallel implementation of A-star pathfinding algorithm.
- Profile GPU kernels using NVIDIA's NCU-CLI, revealing a 90%+ DRAM throughput with memory bound workloads
- Constructed and manipulated unit grid graphs with adjustable obstacle configuration for benchmarking traversal performance.

NHL Game Outcome Predictor: Website — Spring 2024

- Built real-time game prediction model using 1.75M+ NHL shot events from MoneyPuck.com.
- Leveraged machine learning tactics to predict the outcome of NHL games based on historical shot data.
- Engineered shot quality features for logistic regression, then integrated into a LSTM to predict outcomes dynamically.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, CUDA, Flutter, Haskell, Rust, GDScript.

Back-End: MySQL, PostgreSQL, Node.js, Express.js.

Front-End: HTML, CSS, JavaScript, React.js, Vue.js, tailwind.css.

Operating Systems: Unix/Linux, Windows, Mac OSX.

Software: Git, Microsoft Teams, Lists, Planner, Power BI, Word, Excel, Visual Studio Code, IntelliJ, Xcode, Wireshark, Godot.